

3_Plot

September 28, 2025

```
[2]: import streamlit as st
import pandas as pd
import plotly.express as px
import sys
sys.path.append("/Users/a.h.sheikh/Desktop/IND320_Git_Job/IND320_1_Project/
↳IPYNB/utils.ipynb")
%run utils.ipynb
```

2025-09-28 12:38:42.811 WARNING streamlit.runtime.caching.cache_data_api: No runtime found, using MemoryCacheStorageManager

```
[3]: st.set_page_config(page_title="Plot", page_icon=" ", layout="wide")
st.title("Data plot (Plotly)")
```

2025-09-28 12:38:42.946 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 12:38:42.947 WARNING
streamlit.runtime.scriptrunner_utils.script_run_context: Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2025-09-28 12:38:42.993
Warning: to view this Streamlit app on a browser, run it with the following command:

```
streamlit run /Users/a.h.sheikh/.pyenv/versions/ind320/lib/python3.12/site-packages/ipykernel_launcher.py [ARGUMENTS]
```

2025-09-28 12:38:42.993 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
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```
[3]: DeltaGenerator()
```

```
[4]: # Load data (cached in utils.load_data)
df = load_data()
```

2025-09-28 12:38:43.005 No runtime found, using MemoryCacheStorageManager

```
[5]: # Find a date/time-like column
date_cols = [c for c in df.columns if any(k in c.lower() for k in ["date", "time", "datetime", "timestamp"])]
date_col = date_cols[0] if date_cols else None
if date_col is None:
    st.error("No date/time column found - cannot build a time series.")
    st.stop()

[6]: # Ensure datetime dtype
if not pd.api.types.is_datetime64_any_dtype(df[date_col]):
    df[date_col] = pd.to_datetime(df[date_col], errors="coerce")

[7]: # Numeric columns
num_cols = [c for c in df.columns if pd.api.types.is_numeric_dtype(df[c])]
if not num_cols:
    st.error("No numeric columns found to plot.")
    st.stop()

[8]: # Select column(s) to plot
choice = st.selectbox("Select column(s) to plot", ["All columns"] + num_cols, index=0)
```

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2025-09-28 12:38:43.067 Session state does not function when running a script
without `streamlit run`
2025-09-28 12:38:43.068 Thread 'MainThread': missing ScriptRunContext! This
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```
[9]: # Select month range (default: first month)
if "month" in df.columns and df["month"].notna().any():
    months = sorted(df["month"].dropna().unique().tolist())
    default_month = months[0]
    start, end = st.select_slider("Select month(s)", options=months,
    ↪value=(default_month, default_month))
    mask = (df["month"] >= start) & (df["month"] <= end)
    pdf = df.loc[mask].copy()
    subtitle = f"Months: {start} to {end}"
else:
    pdf = df.copy()
    subtitle = "All rows"
```

2025-09-28 12:38:43.081 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.

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```
[10]: # Build Plotly figure
if choice == "All columns":
    long_df = pdf[[date_col] + num_cols].melt(id_vars=date_col,
    ↪var_name="column", value_name="value")
    fig = px.line(long_df, x=date_col, y="value", color="column",
                  title=f"Time series - {subtitle}",
```

```

        labels={date_col: "Date", "value": "Value", "column": "
↪Column"})
else:
    fig = px.line(pdf, x=date_col, y=choice,
                  title=f"Time series - {choice} - {subtitle}",
                  labels={date_col: "Date", choice: "Value"})

```

```
[11]: st.plotly_chart(fig, use_container_width=True)
```

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